

Materia

Seguridad en el Desarrollo de Aplicaciones

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Crear proyecto

```
dotnet new mvc -n VulnerableApp -f net8.0
cd VulnerableApp
dotnet build
```

Instalar paquetes

```
dotnet add package Microsoft.EntityFrameworkCore.SqlServer --version 8.0.8
dotnet add package Microsoft.EntityFrameworkCore.Tools --version 8.0.8
```

Models/User.cs

```
namespace VulnerableApp.Models
{
    public class User
    {
        public int Id { get; set; }
        public string Username { get; set; } = string.Empty;
        public string Password { get; set; } = string.Empty;
        public string Email { get; set; } = string.Empty;
        public decimal Balance { get; set; }
        public DateTime CreatedAt { get; set; }
    }
}
```

Data/AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;

using VulnerableApp.Models;

namespace VulnerableApp.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options) { }
    }
}
```

```
public DbSet<User> Users { get; set; }

protected override void OnModelCreating(ModelBuilder modelBuilder)
{
    modelBuilder.Entity<User>()
        .Property(u => u.Balance)
        .HasColumnType("decimal(18,2)");

    modelBuilder.Entity<User>().HasData(
        new User
        {
            Id = 1,
            Username = "admin",
            Password = "admin",
            Email = "admin@test.com",
            Balance = 1000m,
            CreatedAt = new DateTime(2025, 1, 1)
        },
        new User
        {
            Id = 2,
            Username = "user1",
            Password = "123456",
```

```

        Email = "user@test.com",

        Balance = 500m,

        CreatedAt = new DateTime(2025, 1, 1)

    },

    new User

    {

        Id = 3,

        Username = "user2",

        Password = "password",

        Email = "user2@test.com",

        Balance = 750m,

        CreatedAt = new DateTime(2025, 1, 1)

    }

);

}

}
}

```

appsettings.json

```

{
  "ConnectionStrings": {
    "DefaultConnection":
"Server=(localdb)\\mssqllocaldb;Database=VulnerableDb;Trusted_Connection=True;MultipleActiveResultSets=true;TrustServerCertificate=True"
  },

  "Logging": {
    "LogLevel": {
      "Default": "Information",

```

```
        "Microsoft.AspNetCore": "Warning"
    }
},

    "AllowedHosts": "*"
}
```

Program.cs

```
using Microsoft.EntityFrameworkCore;

using VulnerableApp.Data;

var builder = WebApplication.CreateBuilder(args);

builder.Services.AddControllersWithViews();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));

var app = builder.Build();

if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");

    app.UseHsts();
}
```

```
}

app.UseHttpsRedirection();

app.UseRouting();

app.UseAuthorization();

app.MapStaticAssets();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}")
    .WithStaticAssets();

app.Run();
```

Crear migración

dotnet ef migrations add InitialCreate

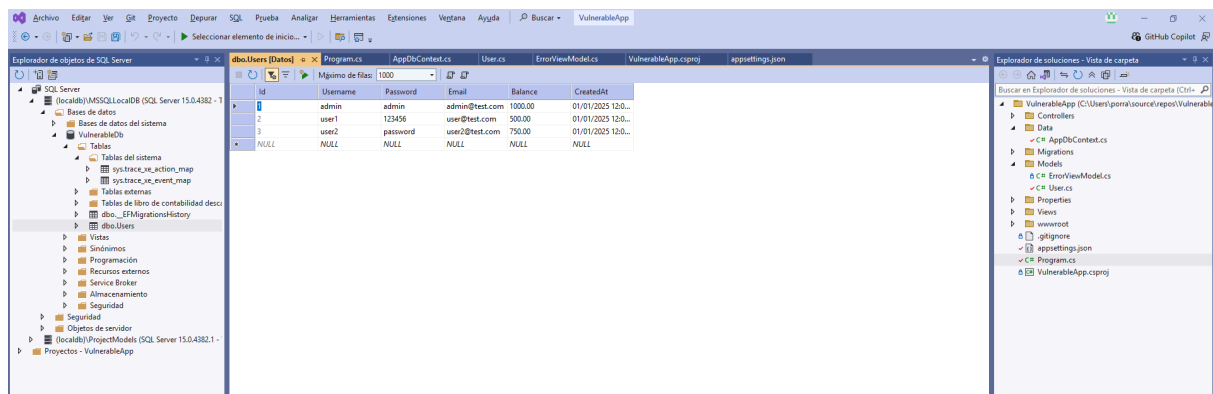
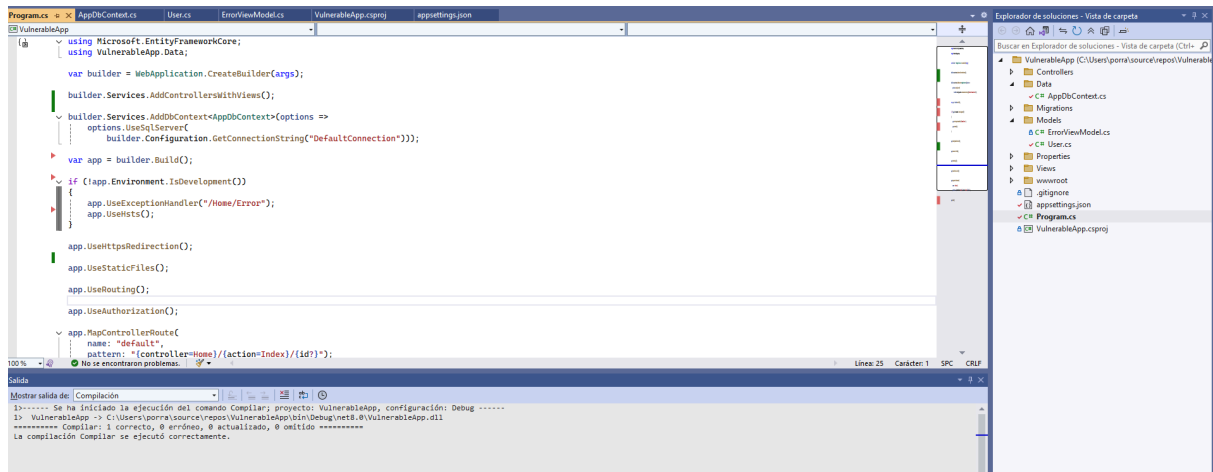
Crear base de datos

dotnet ef database update

Ejecutar proyecto

dotnet run

Evidencias



```

C:\Users\porra\source\repos\VulnerableApp>dotnet ef database update
Build started...
Build succeeded.
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (37ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (17ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT OBJECT_ID(N'[__EFMigrationsHistory]');
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT OBJECT_ID(N'[__EFMigrationsHistory]');
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (4ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT [MigrationId], [ProductVersion]
      FROM [__EFMigrationsHistory]
      ORDER BY [MigrationId];
info: Microsoft.EntityFrameworkCore.Migrations[20405]
      No migrations were applied. The database is already up to date.
No migrations were applied. The database is already up to date.
Done.

```